

TABLE II
Compatible Features of Various Families

Xilinx FPGA Family	Device Database, XDL Parsing, Manipulation and Export	Placement Capabilities	Router Capabilities	Bitstream Parsing, Manipulation and Export
Artix 7	✓	✓		
Kintex 7	✓	✓		
Spartan 2	✓	✓		
Spartan 2E	✓	✓		
Spartan 3	✓	✓		
Spartan 3A	✓	✓		
Spartan 3ADSP	✓	✓		
Spartan 3E	✓	✓		
Spartan 6	✓	✓		
Virtex	✓	✓		
Virtex E	✓	✓		
Virtex 2	✓	✓		
Virtex 2 Pro	✓	✓		
Virtex 4	✓	✓	✓	✓
Virtex 5	✓	✓	✓	✓
Virtex 6	✓	✓	✓	✓
Virtex 7	✓	✓		
Zynq	✓	✓		

kit are on your PATH. Add all the supporting JAR files in the JAR folder (hessian-4.0.6.jar, jopt-simple-3.2.jar, qtjambi-4.6.3.jar and qtjambi-<your_platform>-4.6.3.jar) to your CLASSPATH environment variable. Also add

the RapidSmith Java project to CLASSPATH environment variable. Now you need to create an environment variable RAPIDSMITH_PATH and set the value to the path of Java project located on your computer.

In case you are using the RapidSmith for a legacy Xilinx device such as Spartan 2/E, Virtex or Virtex E/2/2Pro, ISE Design Suite 10.1 or earlier will be required. The program should be pointed to the appropriate installation of tools. For this, we use an environment variable XILINX_LEGACY_PATH and set its value to bin path of the design suite tools. For example, XILINX_LEGACY_PATH=/opt/xilinx/10.1/ISE/bin/win32 will use the 32-bit tools of ISE 10.1 of a Windows installation. Skip this step if you are not using an outdated device.

Next we have to compile all the Java classes. This is done automatically if the project is imported to an IDE—like Eclipse. Now test the installation by running any of the programs.

Overview

RapidSmith is organised into several packages. These packages and their description are listed in Table III. All these packages are prefixed with edu.byu.ece.rapidSmith.

Design analysis tools

RapidSmith provides a simple platform to create design analysis tools for a



Fig. 1: A simple design browser tool

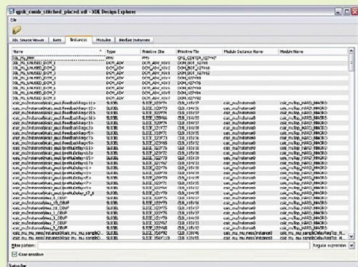


Fig. 2: A program using RapidSmith

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